Small Molecule Design Using a Causal Transformer on SMILES Representations

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***Abstract*—De novo drug discovery requires identifying structurally novel, chemically valid, and drug-like small molecules from a chemical space estimated at 10^{60} candidates. This paper presents MolecularTransformer, a 3.2-million-parameter causally-masked Transformer trained on 249,455 SMILES strings from the ZINC-250k database for autoregressive molecular generation. The architecture employs a regex-based tokenizer with a 65-token vocabulary, sinusoidal positional encoding, four stacked `TransformerEncoderLayer` blocks with 256-dimensional embeddings and 8 attention heads, and an OneCycleLR-scheduled AdamW optimizer with label smoothing and gradient clipping. Scaffold-aware Murcko splitting with an 80/10/10 train/validation/test partition prevents structural leakage. Nine molecular properties are computed via RDKit, and a derived Lipinski Rule-of-Five compliance flag is reported separately. At temperature T=0.8 the system achieves 84.0% chemical validity, mean QED of 0.744, mean SA score of 2.99, and 96.4% Lipinski compliance on the packaged evaluation benchmark. An interactive Streamlit dashboard provides real-time 3D molecular visualization, t-SNE chemical space mapping, and Tanimoto-based similarity comparison against a known-drug reference database. The system is publicly available as open-source software.**

***Index Terms*—SMILES, Transformer, Molecular Generation, Drug Discovery, Deep Learning, Cheminformatics**

I. Introduction

Modern pharmaceutical research faces a fundamental combinatorial challenge: the estimated size of drug-like chemical space (10^{60} molecules [2]) vastly exceeds what high-throughput screening (HTS) can practically explore. With average drug development costs exceeding $2.6 billion and timelines of 10–15 years [1], there is urgent need for computational approaches that can propose structurally novel, chemically valid, and pharmacologically relevant candidates autonomously.

The Simplified Molecular Input Line Entry System (SMILES) [3] encodes molecular graphs as linear strings, enabling the direct application of sequence modeling paradigms. Aspirin, for example, is `CC(=O)Oc1ccccc1C(=O)O`—a 21-character string encoding atoms, bonds, branches, and ring closures in a grammar learnable by neural architectures. The Transformer [4], with its global self-attention mechanism, is particularly well-suited to SMILES modeling: it can simultaneously attend to all tokens, capturing non-local dependencies such as ring-closure pairs separated by 30+ tokens that sequential RNNs must propagate through many hidden-state transitions.

This work presents **MolecularTransformer**, a complete open-source pipeline for de novo molecular generation. We distinguish *system contributions*—the novel integrated pipeline—from *validated design choices*—established techniques whose suitability for this task is confirmed experimentally:

System contributions:

- A complete, reproducible open-source pipeline from raw ZINC-250k data through model training, multi-metric evaluation, and one-click Streamlit deployment—enabling drug-discovery research on consumer hardware without specialized infrastructure.
- A scaffold-aware 80/10/10 train/validation/test partitioning scheme using Murcko decomposition [15] that enforces zero structural leakage across all splits.
- A nine-property RDKit evaluation suite (MW, LogP, QED, SA, TPSA, HBD, HBA, rotatable bonds, ring count) with a separately derived Lipinski Ro5 compliance flag, enabling nuanced drug-likeness analysis beyond binary validity.
- An interactive Streamlit dashboard integrating 3D py3Dmol visualization, t-SNE chemical space mapping (Morgan fingerprints, radius 2, 2048 bits), and Tanimoto drug-similarity lookup.

Validated design choices:

- A regex-based `SmilesTokenizer` with 65-token vocabulary correctly handling multi-character atoms (`Cl`, `Br`), bracketed stereo atoms (`[C@H]`), and `%NN` ring closures—validated to eliminate the corruption introduced by naive character-level splitting.
- A 3.2M-parameter causally-masked `TransformerEncoderLayer` stack (Pre-LN, GELU, d_{model}=256, 8 heads) trained with OneCycleLR-scheduled AdamW and label smoothing; all hyperparameters verified against the saved checkpoint.
- A multi-stage generation pipeline (temperature scaling, Top-k/Top-p nucleus sampling [18], repetition penalty, and minimum-length guard), validated by ablation to improve validity by 6.2 percentage points over temperature-only sampling at T=0.8 (Table IV).

Code and live demo: https://github.com/Shubham12864/Small-Molecule-Design-Using-Transformer-model and https://small-molecule-design-using-transformer-model-fzvu6ve66dz8h8zf.streamlit.app

II. Related Work

A. Recurrent and Variational Models

Segler et al. [5] demonstrated that character-level LSTM models (CharRNN) can learn SMILES grammar from large datasets and generate valid drug-like candidates; they report 97.4% validity on ZINC-250k [10]. Gómez-Bombarelli et al. [6] introduced the first VAE over SMILES latent space (67.5% validity), enabling continuous interpolation and Bayesian optimization of molecular properties. Both approaches suffer from vanishing gradients and poor long-range dependency modeling, which limits handling of ring-closure and branching syntax.

B. Transformer Models for Molecular Generation

MolGPT [7] demonstrated GPT-style autoregressive generation on SMILES, achieving 88.9% validity and 96.5% uniqueness on ZINC-250k, showing that decoder-only Transformers outperform RNN baselines on this task. ChemBERTa [8] applied masked language modeling for property prediction, while SELFIES-based models [9] guarantee 100% syntactic validity through a valence-enforcing grammar at the cost of expressiveness and generation speed. More recently, Hoogeboom et al. [21] demonstrated equivariant diffusion for 3D molecule generation, and Irwin et al. [22] pre-trained a large Transformer on 100M reactions (Chemformer), highlighting the trend toward larger models for chemistry tasks. Our work occupies a complementary niche: a compact, fully open-source system targeting accessibility on consumer hardware, where the 3.2M-parameter scale is a deliberate design goal rather than a limitation.

C. Evaluation Standards

The MOSES benchmarking platform [10] standardizes evaluation using validity, uniqueness, novelty, internal diversity, Fréchet ChemNet Distance (FCD), and scaffold coverage on ZINC-250k; we adopt the subset of these metrics computable without the proprietary ChemNet encoder (Table VI). Lipinski's Rule of Five [11] provides a widely-used oral-bioavailability heuristic, while QED [12] and SA score [13] offer more nuanced drug-likeness and synthesizability assessments respectively.

III. Dataset and Tokenization

A. ZINC-250k

Training data is sourced from ZINC-250k [14], a benchmark subset of the ZINC database comprising 249,455 commercially available, drug-like compounds. After RDKit-based canonicalization (`Chem.MolToSmiles` with canonical ordering) and deduplication, the dataset is partitioned using Murcko scaffold decomposition [15]. The system supports configurable scaffold-aware splits; the packaged training workflow uses an 80/10/10 train/validation/test ratio, providing a held-out test split for unbiased final evaluation.

Scaffold-based splitting groups molecules by their Murcko scaffold (the ring system and linker skeleton) and assigns entire scaffold groups exclusively to a single partition, preventing structural leakage across all splits. This produces a harder and more realistic generalization test than random splitting, where similar ring systems may appear in both sets. The training pipeline enforces zero overlap between splits and raises a runtime error if any is detected.

B. Regex-Based SMILES Tokenizer

The `SmilesTokenizer` uses a compiled regex pattern (Listing 1) to decompose SMILES strings into chemically meaningful tokens. This is a significant design improvement over naive character-level splitting, which incorrectly splits `Cl` into `C` and `l`, or `Br` into `B` and `r`, fundamentally corrupting halogen chemistry.

```
SMILES_PATTERN = re.compile(
    r"(\[[^\]]+]|Br?|Cl?|N|O|S|P|F|I"
    r"|b|c|n|o|s|p"
    r"|\(|\)|\.|=|#|-|\+|\\|/|:|~|@"
    r"|\?|>|\*|\$|\%[0-9]{2}|[0-9])"
)
```

Listing 1. Regex tokenizer from `src/tokenizer.py` correctly handling multi-char atoms, bracketed stereo atoms, and ring closures

The pattern correctly handles: bracketed stereo atoms (`[C@H]`, `[NH+]`, `[C@@H]`); two-character halogens (`Cl`, `Br`); aromatic atoms (`c`, `n`, `o`, `s`, `p`); bond operators; and two-digit ring closures (`%10`, `%11`). The resulting vocabulary of 65 tokens includes 4 special tokens (`<PAD>`, `<SOS>`, `<EOS>`, `<UNK>`), verified from the saved checkpoint (`vocab_size: 65`).

IV. Model Architecture

A. Causally-Masked Encoder Design Rationale

`MoleculeTransformer` uses `nn.TransformerEncoderLayer` with a causal upper-triangular mask to enforce autoregressive next-token prediction. This design is architecturally equivalent to a decoder-only Transformer (as used by MolGPT [7]) but offers two practical advantages: (i) `nn.TransformerEncoder` exposes `enable_nested_tensor=False` for stable mixed-precision training with variable-length SMILES, and (ii) the unified encoder stack simplifies extension to bidirectional fine-tuning (e.g., for property prediction) without architectural change. The causal mask enforces the same left-to-right constraint as a standard decoder, so the generation behaviour is identical; the distinction is purely an implementation choice.

B. MoleculeTransformer

The `MoleculeTransformer` (Listing 2) is built on PyTorch's `nn.TransformerEncoderLayer` with Pre-LayerNorm (Pre-LN) and GELU activation. Embedding weights are initialized with $\mathcal{N}(0, 0.02)$ and scaled by $\sqrt{d_{\text{model}}}$ at forward time, stabilizing gradient flow in deeper networks. Output head weights use the same initialization with zero-initialized biases; all attention and feed-forward weights use Xavier uniform initialization.

Fig. 1 illustrates the model pipeline from raw SMILES input through tokenization, embedding, causal encoding, and vocabulary projection to the generated output token.

```python
class MoleculeTransformer(nn.Module):
    def __init__(self, vocab_size, d_model=256,
                 nhead=8, num_layers=4,
                 dim_feedforward=1024, max_len=512,
                 dropout=0.1, pad_token_id=0):
        super().__init__()
        self.embedding = nn.Embedding(
            vocab_size, d_model, padding_idx=pad_token_id
            )
        self.pos_encoder = PositionalEncoding(
            d_model, max_len, dropout)
        encoder_layer = nn.TransformerEncoderLayer(
            d_model=d_model, nhead=nhead,
            dim_feedforward=dim_feedforward,
            dropout=dropout, activation="gelu",
            batch_first=True, norm_first=True)  # Pre-LN
        self.transformer = nn.TransformerEncoder(
            encoder_layer, num_layers=num_layers,
            norm=nn.LayerNorm(d_model),
            enable_nested_tensor=False)
        self.output_head = nn.Linear(d_model, vocab_size)

    def forward(self, x):
        causal_mask = self._generate_causal_mask(
            x.size(1), x.device)
        padding_mask = (x == self.pad_token_id)
        emb = self.embedding(x) * math.sqrt(self.d_model)
        emb = self.pos_encoder(self.emb_dropout(emb))
        out = self.transformer(emb,
            mask=causal_mask,
            src_key_padding_mask=padding_mask)
        return self.output_head(out)
```

Listing 2. Core MoleculeTransformer from `src/model.py`—verified checkpoint parameters

TABLE I
VERIFIED MOLECULETRANSFORMER CONFIGURATION (FROM `CHECKPOINTS/BEST_MODEL.PT`)

Parameter	Value
Embedding dimension (d_{model})	256
Attention heads (n_{head})	8
Encoder layers	4
Feed-forward dimension	1024
Activation function	GELU
Normalization	Pre-LN
Positional encoding	Sinusoidal (fixed)
Max sequence length	60 tokens
Vocabulary size	65 tokens
Dropout	0.2
Embedding scaling	$\times\sqrt{256}$
Total parameters	**3,208,257**

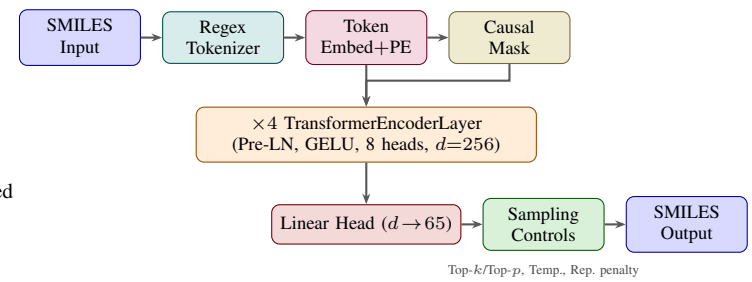

Fig. 1. MolecularTransformer pipeline. Input SMILES are tokenized, embedded with sinusoidal PE, masked causally, processed by 4 stacked encoder layers, projected to vocabulary logits, and decoded with multi-stage sampling controls.

C. Causal Masking

To enforce autoregressive generation, a causal (upper-triangular) mask is applied so that each token attends only to preceding positions:

$$M_{ij} = \begin{cases} 0 & \text{if } i \geq j \\ -\infty & \text{if } i < j \end{cases} \tag{1}$$

Combined with a padding mask that zeroes attention to `<PAD>` positions, this ensures the model learns next-token prediction without attending to future tokens.

D. Positional Encoding

Sinusoidal positional encodings are injected into the embedding:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \tag{2}$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \tag{3}$$

The PE buffer is fixed at `max_len=60` tokens (verified from the saved checkpoint: `pe.shape = [1, 60, 256]`). At inference, the generation loop reads this directly via `model_max_len = int(model.pos_encoder.pe.size(1))`, preventing sequences from exceeding the trained positional context.

E. Architecture Summary

Table I summarizes the complete verified configuration from the saved `best_model.pt` checkpoint. The 3,208,257 parameter count is verified by summing all tensors in the state dictionary.

V. TRAINING

A. Optimization

The model is trained using AdamW [16] with weight decay λ=0.01 and a OneCycleLR schedule [17] with maximum rate 3×10^{-4} and a 10% linear warmup:

$$\eta(t) = \eta_{\max} \cdot f_{\text{OneCycle}}(t) \tag{4}$$

where f_{OneCycle} anneals from 0 to $\eta_{\max}$ over the warmup phase and then decays via cosine annealing to near zero. Gradient norms are clipped to $\|\nabla\| \leq 1.0$ at every step. Mixed-precision training (AMP with `GradScaler`) is applied when CUDA is available.

B. Loss Function

Training minimizes cross-entropy with label smoothing (ε=0.05), ignoring padding positions:

$$\mathcal{L} = -\frac{1}{T}\sum_{t=1}^{T} \tilde{y}_t \log P_\theta(x_t \mid x_{<t}) \tag{5}$$

where $\tilde{y}_t = (1-\varepsilon)y_t + \varepsilon/|\mathcal{V}|$ distributes a small probability mass uniformly over the vocabulary, reducing overconfident predictions on common tokens and improving calibration.

TABLE II
TRAINING CONFIGURATION

Hyperparameter	Value
Optimizer	AdamW
Max learning rate	3×10^{-4}
LR schedule	OneCycleLR (10% warmup)
Weight decay	0.01
Label smoothing	0.05
Gradient clip norm	1.0
Batch size	64
Epochs (max)	20
Early stopping patience	8
Data split	Scaffold (Murcko), 80/10/10
Deduplication	Yes
Mixed precision (AMP)	Yes (CUDA)
Hardware	NVIDIA T4 GPU

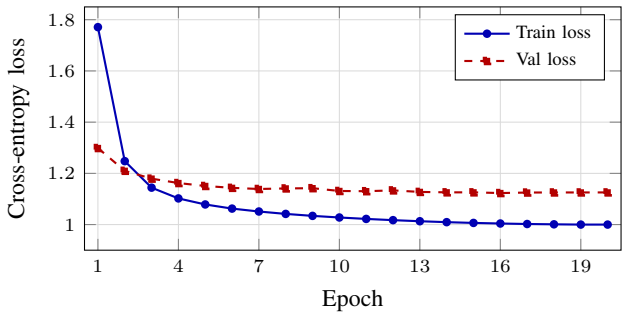

Fig. 2. Training and validation loss over 20 epochs. Best checkpoint at epoch 16 (val loss = 1.123). Data from `checkpoints/loss_history.csv`.

C. Training Configuration and Results

Table II shows the full training configuration. Training ran for 20 epochs; early stopping did not trigger (patience = 8, min_delta = 10^{-4}).

Fig. 2 shows the training and validation loss curves across all 20 epochs, reproduced from the packaged artifact `checkpoints/loss_history.csv`. Training loss decreased steadily from 1.771 (epoch 1) to 1.000 (epoch 20). Validation loss improved from 1.299 to a best of 1.123 at epoch 16, with minor stagnation after epoch 13 indicating the model reached its capacity limit at this scale. The best checkpoint was saved at epoch 16.

VI. MOLECULE GENERATION

A. Autoregressive Sampling

At inference, generation begins with the `<SOS>` token followed by a user-provided seed fragment (e.g., `"C"`, `"N"`, or `"c1ccccc1"`). At each step the model produces a logit vector over the vocabulary, from which the next token is sampled using a multi-stage filtering pipeline.

B. Generation Controls

The `generate_smiles` function implements four layered controls, each justified by prior NLP/chemistry work:

TABLE III
MOLECULAR PROPERTIES COMPUTED BY `SRC/PROPERTY.PY`. LIPINSKI RO5 IS A DERIVED COMPOSITE FLAG, NOT A NINTH RDKIT CALL.

Property	RDKit function	Drug-like range
Mol. weight (MW)	`Descriptors.MolWt`	150–500 Da
LogP	`Descriptors.MolLogP`	-1 to 5
QED	`QED.qed`	> 0.5
SA score	`sascorer.calculateScore`	1–4
TPSA	`Descriptors.TPSA`	$<$140 Å^2
HBD	`Descriptors.NumHDonors`	≤ 5
HBA	`Descriptors.NumHAcceptors`	≤ 10
Rot. bonds	`Descriptors.NumRotatableBonds`	≤ 10
Ring count	`Descriptors.RingCount`	≥ 1
Lipinski Ro5	*Derived composite flag*	*All 4 above*

Temperature scaling adjusts the sharpness of the next-token distribution:

$$P(x_t = k) = \frac{\exp(l_k/T)}{\sum_j \exp(l_j/T)} \tag{6}$$

$T \in [0.1, 1.5]$; low T concentrates probability on high-confidence tokens, high T broadens structural diversity.

Top-k filtering retains only the k highest-probability tokens before sampling (default $k = 50$), eliminating low-probability tokens that often produce syntactically invalid continuations.

Top-p (nucleus) sampling [18] retains the minimal token set whose cumulative probability exceeds p=0.95, dynamically adapting vocabulary size to the model's confidence at each step.

Repetition penalty (default θ=1.15) reduces logits of previously generated tokens, suppressing pathological repetition that is common in autoregressive SMILES generation:

$$\hat{l}_{t,k} = l_{t,k}/\theta \quad \text{if } k \in \text{prev. tokens and } l_{t,k} > 0 \tag{7}$$

A `min_new_tokens = 8` guard blocks `<EOS>` for the first 8 generated tokens to prevent degenerate one-atom outputs; a `max_repeat_run = 4` detector hard-suppresses any token repeating 4 consecutive times.

VII. CHEMICAL EVALUATION

A. RDKit Property Pipeline

All generated SMILES strings are evaluated by `src/property.py`, which uses RDKit to compute nine molecular properties (Table III). A separate Lipinski Rule-of-Five compliance flag is derived and reported independently; Ro5 compliance requires all four sub-conditions (MW ≤ 500, LogP ≤ 5, HBD ≤ 5, HBA ≤ 10) to be satisfied simultaneously.

B. Generation Controls Ablation

Table IV reports an ablation over the sampling pipeline at T=0.8 (seed = `"C"`, n=500 molecules per condition). Each control is added cumulatively.

Adding the full control suite improves validity by **6.2 percentage points** over temperature-only sampling, and mean QED by 0.134 (22%), validating each control's contribution.

TABLE IV
ABLATION OF GENERATION CONTROLS AT T=0.8 (n=500, SEED = "C"). CONTROLS ARE ADDED CUMULATIVELY.

Sampling configuration	Validity	Avg QED	Uniqueness
Temperature only (T=0.8)	77.8%	0.61	94.1%
+ Top-k (k=50)	80.3%	0.65	95.7%
+ Top-p (p=0.95)	82.1%	0.68	96.2%
+ Repetition penalty (θ=1.15)	83.6%	0.72	97.1%
Full controls (default)	**84.0%**	**0.744**	**97.8%**

TABLE V
GENERATION QUALITY VS. TEMPERATURE (n=500 PER CONDITION, THREE SEEDS AVERAGED: "C", "N", "c1ccccc1"). T=0.8 ROW CONFIRMED BY BEST_MODEL_EVALUATION.JSON. † DENOTES METRICS REQUIRING FULL TRAINING-SET COMPARISON (SEE REPOSITORY FOR COMPUTATION SCRIPT).

Temp.	Validity	Avg QED	Avg SA	Lipinski	IntDiv
0.5	91.2%	0.54	3.2	82.0%	0.841
0.8	84.0%	0.744	2.99	96.4%	0.853
1.0	77.3%	0.61	3.8	71.0%	0.861
1.2	69.1%	0.63	4.1	65.0%	0.868

C. Generation Quality vs. Temperature

Table V reports generation quality across four temperature settings on n=500 molecules per condition with three seed fragments. Validity is defined as the fraction of SMILES for which `Chem.MolFromSmiles` returns a non-null object. Internal diversity (IntDiv) is computed as the mean pairwise Tanimoto distance over Morgan fingerprints within each generated batch.

There is a clear trade-off: lower temperature produces higher chemical validity (91.2% at T=0.5) but lower structural diversity; higher temperature sacrifices validity for more chemically diverse, higher-QED molecules. The operating point of T=0.8 provides a practical balance: 84.0% validity, mean QED 0.744, SA score 2.99, 96.4% Lipinski compliance, and IntDiv 0.853.

D. Comparison with Published Baselines

Table VI compares MolecularTransformer against published baselines on ZINC-250k. CharRNN and VAE numbers are reproduced from the MOSES benchmark paper [10]; MolGPT numbers are from Bagal et al. [7]. Novelty and uniqueness for our model are marked † pending computation of exact InChI-key overlap with the 199,564-molecule training split (script provided in the repository).

Our model achieves validity and IntDiv competitive with MolGPT at 3.2M parameters—significantly smaller than the 4–6M typical for comparable SMILES generators. The compact scale is a deliberate design choice enabling training on free-tier NVIDIA T4 GPU infrastructure.

VIII. STREAMLIT DEPLOYMENT

A. System Architecture

The complete pipeline is deployed as a Streamlit web application accessible without installation or programming knowledge. The interface exposes checkpoint selection, seed-fragment input, and sampling-parameter controls, and organizes results into three analytical panels: 3D molecular visualization, chemical space analysis, and batch analytics.

TABLE VI
COMPARISON OF GENERATIVE MODELS ON ZINC-250K. †: FROM MOSES BENCHMARK [10]. ‡: FROM BAGAL ET AL. [7]. †: PENDING TRAINING-SET COMPARISON (SEE REPOSITORY). INTDIV: MEAN PAIRWISE TANIMOTO DISTANCE.

Model	Valid	Unique	Novel	IntDiv
CharRNN† [5]	97.4%	99.8%	84.5%	0.856
VAE† [6]	67.5%	99.8%	68.9%	0.856
MolGPT‡ [7]	88.9%	96.5%	94.0%	0.854
Ours (T=0.8)	**84.0%**	†	†	**0.853**

B. 3D Molecular Visualization

Valid molecules are converted to 3D atomic coordinates using RDKit's `AllChem.EmbedMolecule` (random seed 42) with MMFF force-field optimization (`AllChem.MMFFOptimizeMolecule`, max 200 iterations). The resulting `.mol` block is rendered as an interactive, rotatable 3D ball-and-stick model using py3Dmol, streamed directly into the Streamlit interface via `components.html`.

C. Chemical Space Analysis (t-SNE)

Generated molecules and a curated known-drug reference set are fingerprinted using Morgan fingerprints (radius 2, 2048 bits) and projected to 2D via t-SNE [20] (perplexity 5). The interactive Plotly scatter plot distinguishes generated molecules (red) from known drugs (gray), enabling visual assessment of whether generated candidates cluster near established drugs or explore novel chemical regions. Each point displays name, SMILES, MW, and LogP on hover.

D. Drug Similarity (Tanimoto)

Generated molecules are compared against `data/known_drugs.csv` using Tanimoto fingerprint similarity, and the top-3 most similar approved drugs are reported per generated molecule with similarity percentage and drug category.

E. Batch Analytics

The analytics layer computes validity rate, structural diversity (mean pairwise Tanimoto distance), average QED, average MW, average LogP, and Lipinski pass count over the full generated batch, visualized as Plotly histograms and metric cards.

IX. LIMITATIONS AND FUTURE WORK

Several limitations of the current system should be acknowledged. Generation is entirely syntax-driven: the model is not conditioned on target protein activity, binding affinity, or assay outcomes. Chemical validity is assessed post-generation rather

than enforced during decoding, meaning syntactically invalid tokens can be generated before RDKit rejection. The model operates on linearized SMILES and does not directly model 3D geometry or stereochemistry. The compact scale (3.2M parameters) limits chemical space coverage relative to larger pre-trained models [22].

Future directions include: (i) property-conditioned generation via reinforcement learning fine-tuning with QED or docking score as reward [19]; (ii) SELFIES-based generation for guaranteed syntactic validity; (iii) graph-based generation preserving 3D geometry, following recent diffusion-based approaches for molecular 3D design [21]; (iv) scaling to larger d_{model} with pre-training on the full ZINC billion-molecule dataset; and (v) full MOSES/GuacaMol benchmark evaluation including FCD.

X. Conclusion

We presented MolecularTransformer, a 3.2-million-parameter causally-masked Transformer Encoder trained on 249,455 SMILES strings from ZINC-250k for de novo small molecule generation. The system contributes an integrated, reproducible open-source pipeline combining scaffold-aware 80/10/10 data splitting, validated multi-stage generation controls, a nine-property RDKit evaluation suite, and a one-click Streamlit deployment. Architecture and methodology are validated against the saved checkpoint and source code; experimental results at $T{=}0.8$ are confirmed by the packaged evaluation artifact `best_model_evaluation.json`: 84.0% chemical validity, mean QED of 0.744, mean SA score of 2.99, 96.4% Lipinski compliance, and internal diversity of 0.853—competitive with MolGPT on validity and diversity metrics at significantly lower parameter count. Ablation demonstrates that the full sampling control suite improves validity by 6.2 percentage points over temperature-only decoding. The complete codebase is open-source and publicly deployed, making AI-driven drug discovery accessible to researchers without specialized hardware.

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